

The background features a network diagram of interconnected nodes and lines in a light orange color, set against a darker orange gradient. On the left side, there are several diagonal stripes in shades of gray, light blue, and orange. The title text is positioned in the upper right area of the slide.

Introduction to Internet and WWW

Internet

Internet stands of “**international networks**”. It is a network of millions of computers connected to each other worldwide, that provides information on almost every subject and facilitates data transmission and exchange of data resources.

In **India** internet services started on **15 Aug 1995** through **VSNL** (Videsh Sanchar Nigam Limited)

Private internet service provider are – Airtel, Reliance, Sify, Tata etc.

Uses of internet

- Education
- Email
- Business
- E-commerce
- Media and Entertainment
- Social networking
- Health and fitness
- Forum

What is a Computer Network

- A computer network is a group of computers, and peripheral devices connected through a data communication system.
- It facilitates the sharing of hardware and software resources among different users working on different computers.
- These shared resources can be data files, any application software or hardware devices like printer modem. etc.

Advantage of network

Hardware & Software sharing:-

In a computer network, data, s/w, h/w resources can be shared irrespective of the physical location of the resources and the users. All these devices are attached to the server, which is shared among number of users.

Reliability:-

A file can have its copies on two or more computer of the network, so if one of them is unavailable, the other copies could be used. That makes network more reliable.

Reduced cost:-

In a network, we can share almost every peripheral device, such as printer scanner, modem etc. among several computers. This reduce the cost of hardware.

Fast communication:-

With network, it is possible to exchange information at very fast speeds.

Evolution of Networking

ARPANET :-

The first network was developed in 1969, when the US Defense Dept. sponsored a project named **ARPANET** (ADVANCE RESEARCH PROJECT AGENCY NETWORK). It was developed by **Robert Kahn** And **Vint Cerf**

NSFnet:-

In 1980, **NSFnet** (National Science Foundation) was started to make a high-capacity network to be used strictly for academic and engineering research. It is more capable than ARPANET.

Internet:-

In 1990, the internetworking of ARPANET , NSFnet and other private networks resulted into internet.

Components of Computer Network

There are many components of the network, which share common devices, functions & features including

Server

A server provide services to other computers, which holds stored files, programs and network operating system.

Client

The client computer access network & shared resources provided by the server. It is less powerful than server computer.

Transmission Media

Transmission media interconnect the computers in a network.

Different types of transmission media are, Wired or Bounded and unguided or wireless.

Components of Computer Network

Network Interface Card (NIC):

It is also known as adopter. NIC regulates the exchange of data between the network and the computers.

Protocols:

A set of rules is called protocol. network protocol is standardized set of rules used for transmission of data and/or establish communication between them.

Resources:

The hardware that are available to a computer on a particular network are known as resources.

Network Devices

Network devices are required to amplify the signal to restore the original strength of signal and to provide an interface to connect multiple computer in a network. There are many types of network devices used in networking

Repeater

A repeater is a device that operates only on the physical layer of OSI model. Repeaters have two ports and can connect two segments of LAN. It amplifies the signals, when they are transported over a long distance so that the signal can be as strong as the original signal. A repeater boosts the signal back to its correct level.

Bridge

A bridge is a device which filter data traffic at a network boundary. Bridge reduce the amount of traffic on a LAN by dividing it into two segments. It operates at the data link layer of the OSI model. Traditional bridges support one network boundary, whereas switches usually offer four or more ports.

Network Devices

Hub

Hub is like a repeater with multiple ports used to connect the network channels. It acts as a centralized connection to many computers with the central node or server. When a hub receives a packet of data at one of its port from a network channel, it transmits the packet to all of its ports to all other network channels.

Gateway

A gateway is an interconnecting device, which joins two different network protocols together. They are also known as protocol converters. It accepts packet formatted for one protocol and converts the formatted packet into another protocol.

Router

Router is a hardware device which is designed to take incoming packets, analyze the packets, moving the packets to another network, converting the packets to another network interface, dropping the packets, directing packets to the desired locations, etc.

Switch

It is a small hardware device that joins multiple computer together within one LAN. Switches work on the data link layer of the OSI model. It help to reduce overall network traffic. A switch forwards each incoming packet to the specified recipient. Switches are sometimes called multiport bridges.

Modem

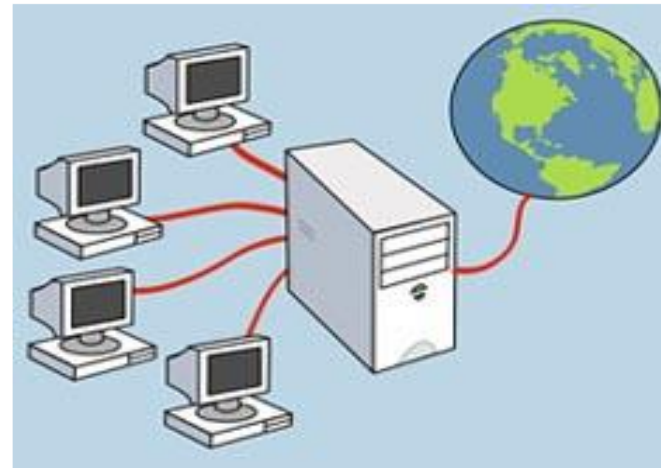
Modem is a device that converts digital signal to analog signal (modulator) at the sender's end and converts back analog signal to digital signal (demodulator) at the receiver's end, in order to make communication possible via telephone lines. A modem is always placed between a telephone line and a computer.

Types of computer networks

- LAN(Local Area Network)
- MAN(Metropolitan Area Network)
- WAN(Wide Area Network)
- PAN(Personal Area Network)

Local Area Network (lan)

LAN is used when two or more peripheral devices are connected within a small area, such as a room, office building or a campus. Maximum range of LAN can be 1KM. The number of computer in a LAN vary from 2 to 1000. the h/w as well as s/w recourses are shared through LAN.



Metropolitan area network (man)

It is much larger than a LAN setup and is spread across a city. Since it can cover an entire city, it is called MAN. The cable television network is the best example of MAN. The main purpose of MAN is also to share h/w and s/w resources. The maximum range of MAN can be 50km.



Wide Area Network(wan)

WAN is used when we have two or more computers located at far away places, linked by communication facilities, like telecommunication or satellite signals. The most common example of WAN is **mobile phone network**. **Internet** is also a WAN. WANs use technology like **ATM, Frame Relay** and **X .25** for connectivity.



Personal area network(pan)

This type of network is mainly created for an individual person. It consists of computer , a mobile etc. you can use these networks to transfer files including email, photo, music etc. PAN can either be wired or wireless. The range of PAN is 10 meters (about 30 feet).

USB and FireWire often link

Together a wired PAN, while a

Wireless PAN typically uses

Bluetooth ,or sometimes infrared

Connections. Bluetooth PAN are

Also called **PICONETS**

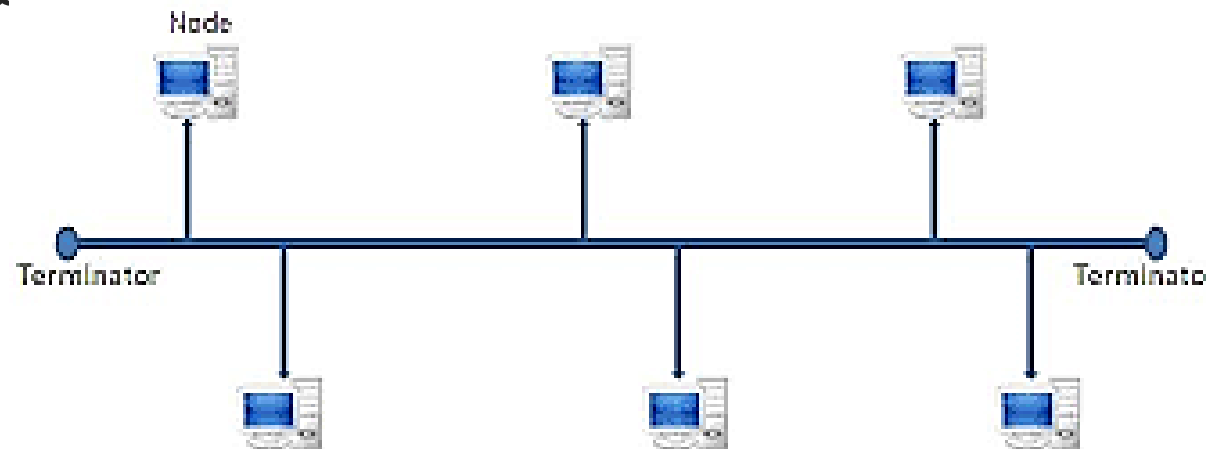


Network topology

The term 'topology' refers to the way a network is laid out, either physically or logically. Topology can be referred as the geometric arrangement of a computer system. Each computer in a topology is known as **node**. The most commonly used topology are:

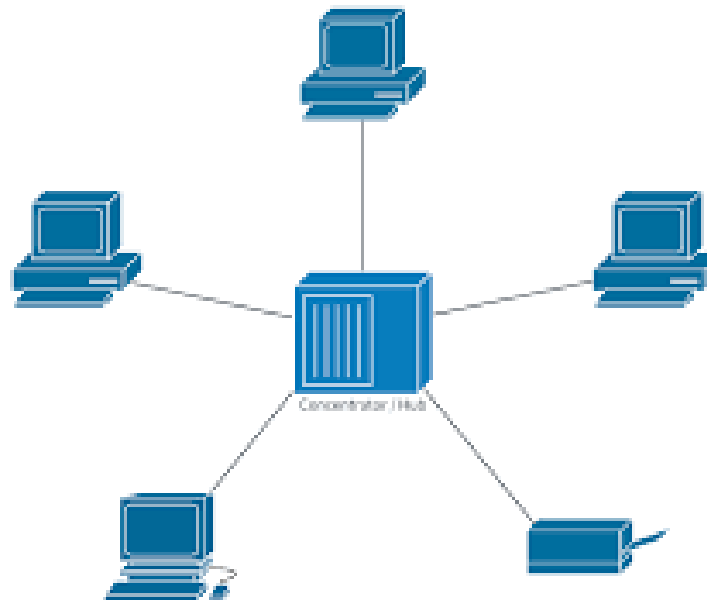
Bus Topology

In Bus topology all network component are connected with single cable in same line called **backbone** in this network topology.



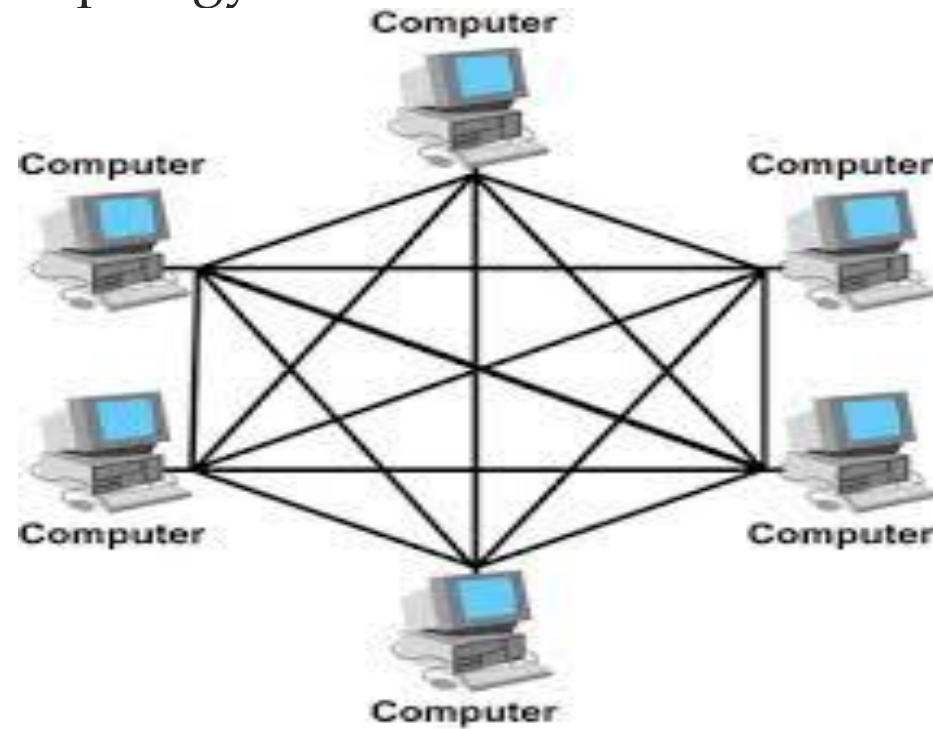
Star Topology

In this network topology, the peripheral nodes are connected to a central node, which rebroadcasts all transmissions received from any peripheral node to all peripheral node across the network. The **protocols** used in star topology are **Ethernet**, **Token Ring** and **Local Talk**



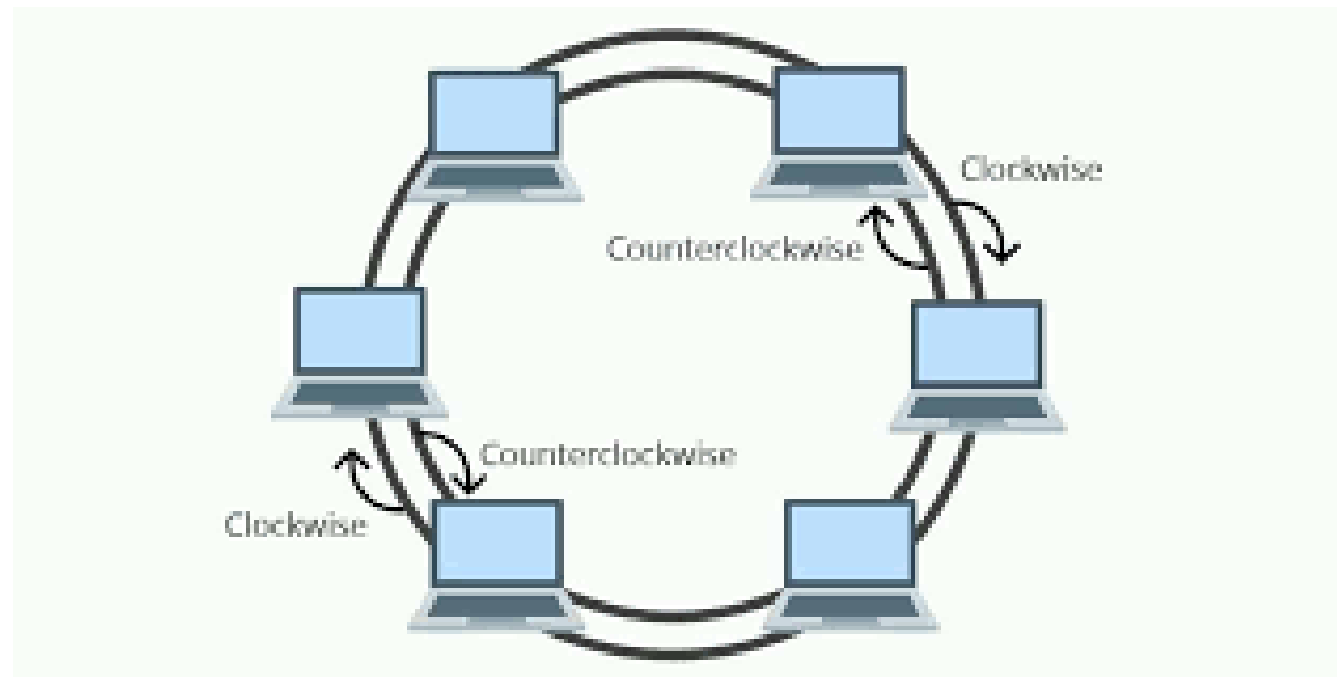
Mesh Topology

In mesh topology, every node has a dedicated Point-to-Point link to every other node. It provides a **bi-directional** link between each possible node. It is also known as completely interconnected topology.



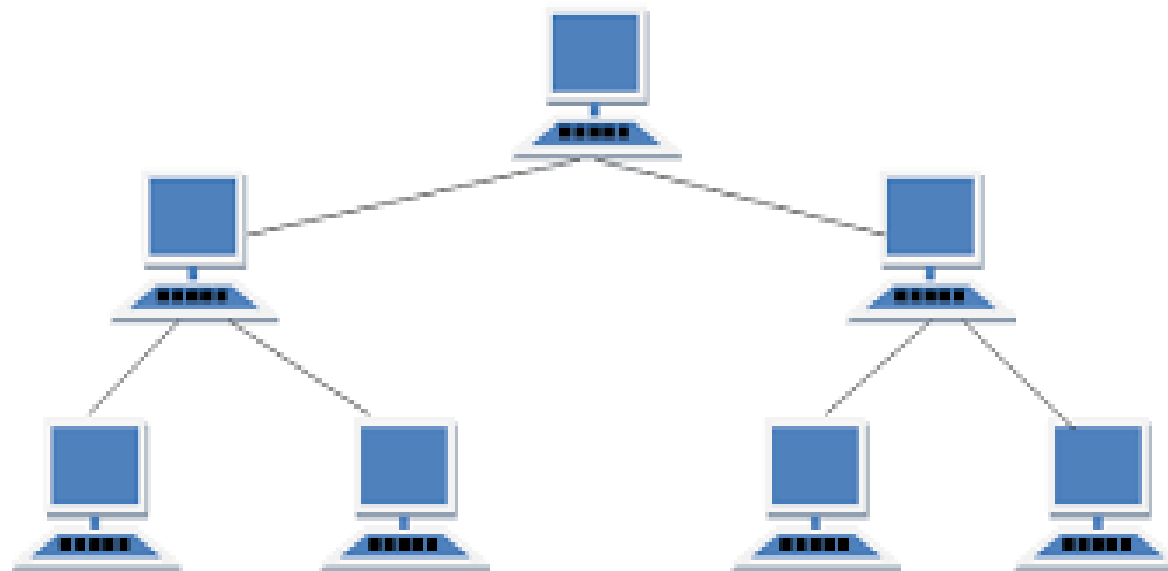
Ring Topology

Ring Topology is used in high performance networks where large bandwidth is necessary. In ring topology, data is transmitted in form of token over a network. The protocols used to implement ring topology are **Token Ring** and **Fiber Distributed Data Interface (FDDI)**



Tree Topology

This is a network topology in which nodes are arranged as a tree. The function of central node in this topology may be distributed. It allows more devices to be attached to a single hub.



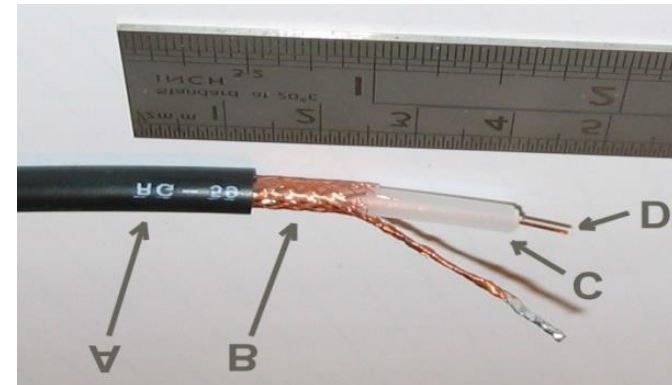
Transmission Media

- Two main categories:
 - **Guided** — wires, cables
 - Twisted-Pair cables:
 - Unshielded Twisted-Pair (UTP) cables
 - Shielded Twisted-Pair (STP) cables
 - Coaxial cables
 - Fiber-optic cables
 - **Unguided** — wireless transmission, e.g. radio, microwave, infrared, sound

Guided media/wired technology

In this data signals are bounded by cabling system which guides the signal along a specific path.

- **Coaxial Cable:-** Coaxial cables are high-frequency transmission cables made up of a single solid-copper core. Data is transferred electrically over the inner conductor and has **80X more transmission capacity** than twisted pair cables.. It is primarily used by cable TV companies to connect their satellite antenna facilities to customer homes and businesses.



Twisted-Pair Cable

It is a light weight easy to install, inexpensive and twisted together two sets of copper wire, surrounded by an insulating material. One wire is used to carry signals to receiver and other is used for ground reference.

There are **two types** of twisted pair cables:

A **Shielded Twisted Pair** (STP) has a fine **wire** mesh surrounding the wires to protect the transmission; an **Unshielded Twisted Pair**(UTP) do not. Shielded **cable** is used in older telephone **networks**, as well as **network** and data communications to reduce outside interference



Fiber optic cable

It allow transmission over longer distance at higher bandwidth. It consists of narrow strand of glass of glass called core, surrounded by concentric layer of glass called cladding. Cladding is further covered by a protective coating of plastic called jacket.



Unguided media

It is the transfer of information over a distance without the use of enhanced electrical wires or conductors. When the computers in a network are interconnected and data is transmitted through waves, then they are said to be connected through unguided media. Some commonly used unguided media of transmission are:

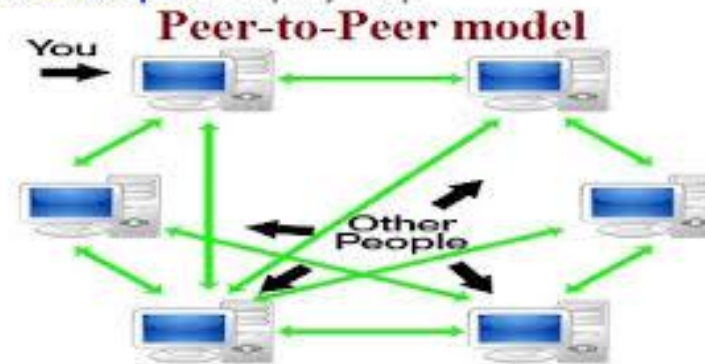
- Radiowave Transmission
- Infrared wave Transmission
- Microwave Transmission
- Satellite Communication

Models of computer networking

Peer to Peer Network

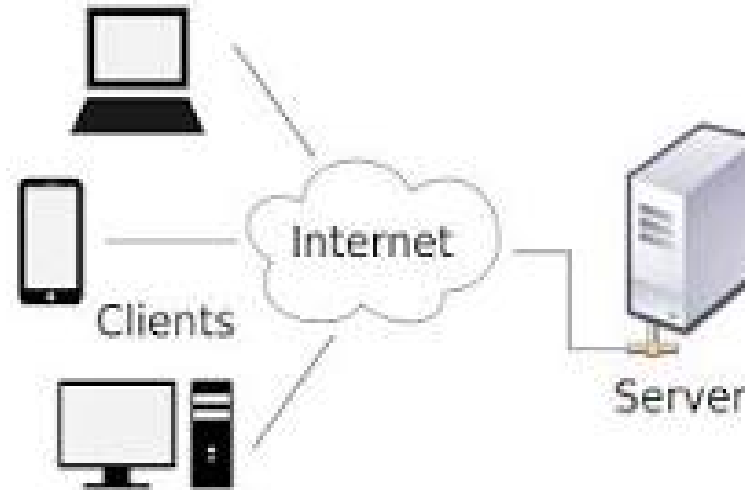
It is a very small network and commonly known as P2P network. Each workstation has equivalent capabilities and responsibilities. It Does not require a switch or a hub. In P2P connection , a couple of computer is connected via Universal Serial Bus (USB) to transfer files like audio, video, data or anything in digital format.

P2P Network Fundamental Concepts Explained
with Example - Step by Step



Client –Server Network

A computer **network** in which one centralized, powerful computer (called the **server**) is a hub to which many less powerful personal computers or workstations (called **clients**) are connected. The **clients** run programs and access data that are stored on the **server**. Here client act as the active device and server behaves as passive device.



Basic of internet architecture

Internet is a global system of interconnected networks. It consists of multiple networks in which LANs are attached to other LANs, other communication networks, remote sites, individual stations and WANs. The architecture of internet is hierarchical in nature.

Services on internet

Services on internet can be divided in following three parts:

1. WWW and Website

WWW and its related term are as following:

World Wide Web(WWW)

The World Wide Web (WWW) is a network of online content that is formatted in HTML and accessed via HTTP. The term refers to all the interlinked HTML pages that can be accessed over the Internet. The World Wide Web was originally designed in **1991** by **Tim Berners-Lee** while he was a contractor at **CERN**.

The World Wide Web is most often referred to simply as "the **Web**."

Web Page

The backbone of the **www** is made of files, called pages or web pages, containing information and links to resources both **text** and **multimedia** throughout the internet. It is created using **HTML** (Hyper Text Markup Language).

Website

A website is a collection of digital documents, primarily HTML files and multimedia files, that are linked together and that exist on the internet under the same domain. In other word, A group of web pages that follow the same theme and are connected together with hyperlinks is called a website. The first page of website is called **home page** of that wesite.

e.g. <https://www.cccguru.com> is a website while <https://www.cccguru.com/practice-set/> is a web page

Web Browser

Web browser is a application s/w that is used to locate, retrieve and display content on the **www**, including web pages. We can install more than one browser on a single computer. User can navigate through files, folders and websites with the help of a browser.

Web Server

A web server is a system that delivers content or services to end users over the internet. A web server consists of a physical server, server operating system (OS) and software used to facilitate HTTP communication. A web server is also known as an **internet server**. The basic **function** of a **web server** is to host websites and to deliver **web** content from its hosted websites over the **internet**. During the delivery of **web** pages, **web servers** follow a network protocol known as hyper text transfer protocol (HTTP).

Web Address/URL

A **website address**, also known as a **URL** (uniform resource locator), is an **Internet** or **Intranet** name that points to a location where a file, directory or **website page** is hosted. They can also be embedded into the code of **web pages** in the form of hyperlinks to direct the user to other locations on the **Internet**.

e.g. <https://www.google.com/services/index.htm>

Domain Name

Domain is a group of network resources assigned to a group of users. A domain name is a way to identify and locate computers connected to the internet. A domain name must be unique. It is always have two or more parts, separated by period(dots). e.g. **google.com**, etc.

Domain Abbreviation

Domains are organized by the type of organizations and by country. A three letter abbreviation indicating the organisation and usually two letter abbreviation indicating country name. most common domains are listed below:

❖ .info (information organisation)	❖ .in (India)
❖ .com (Commercial)	❖ .us (United State)
❖ .gov (Government)	❖ .au (Australia)
❖ .edu (Education)	❖ .fr (France)
❖ .net (Network resources)	❖ Etc.

Domain Name System (DNS)

The Domain Name System stores and associates many type of information with domain names, but most importantly, it translates domain names (computer host names) to IP addresses. It also lists mail exchange servers accepting E-mail for each domain. DNS is an essential component of internet use. The naming scheme by which servers are identified is known as the domain name system.

Number Addressing System (IP Address)

An IP address consists of unique set of numbers provided to each computer by IETF. The IP address is a 32 bit numeric address, containing a set of four numbers, varying from 0 to 255 and separated by periods. The IP address can be directly used to access the web page instead of typing the domain name.

e.g. 205.198.0.1

132.45.10.6

2. Communication on internet

The following services are used for communication on internet:

Blogs

A blog is a website or web page in which an individual records his/her opinions, links to other sites and other data on regular basis. A blog combines text, images and links to other blogs, web pages and other media related to its topics. The entries of blog are known as **posts**.

Newsgroups

It is a online discussion forum on a specified topic, interact through electronic bulletin board system and chat sessions. User post message to a news server which sends them to a bunch of other participating servers, so that other user can access the newsgroup and read the postings.

Chat

Chatting is the online textual or multimedia conversation. It is widely interactive text-based communication process that takes place over the internet.

E-mail (Electronic mail)

E-mail is an electronic version of sending and receiving letters. E-mail lets you send and receive messages in electronic form.

3.Other services on Internet

Apart from communication , there are many other services that are beneficial for consumers.

Video Conferencing

Video conferencing is a communication technology that integrates audio and video data to connect users anywhere in the world as if they are in the same room.

E-Learning (Electronic Learning)

E-learning refers to the electronic mode of delivering, learning, training or educational programs to users. It is the mode of acquiring knowledge by means of the internet and computer based training programs.

E-Banking (Electronic Banking)

E-banking is defined as the automated delivery of new and traditional banking products and services directly to customers through electronic, interactive communication channels. E-banking is also known as internet banking or online banking.

E-Shopping (Electronic Shopping)

E-shopping or online shopping is the process of buying goods and services from merchants (companies) who sell on internet.

e.g. Flipkart, Naaptol etc.

E-Reservation (Electronic Reservation)

E-Reservation means making a reservation for a service via internet.

E-Commerce (Electronic Commerce)

E-commerce includes sharing business information, maintaining business relationship and conducting business transactions by means of telecommunication network.

M-Commerce (Electronic Commerce)

M-Commerce provides the application for buying and selling goods through wireless internet enabled handheld devices. It was started by **Kevin Duffey in 1997**.

Social Networking

A social networking services is an online services or platform that focus on facilitating the building of social networks or social relation among people. Some site is like **Facebook, MySpace etc.**

Preparing computer for internet access

To access internet on a computer, you need following:

ISPs and Examples

ISP (Internet Service Provider) refers to the company that provides internet connections to the users, usually for a fee. The service provider usually provides a software package, username, password and access phone number.

Internet access techniques

Some of the internet connections available for internet access are:

Dial-Up Connection

A Dial-up is a method of connecting to the internet using an existing telephone. The modem connects the computer through the standard phone lines, which serve as the data transfer medium. When a user initiate a dial-up connection, the modem dial a phone number of an ISP that designated to receive dial-up calls. The ISP then establishes the connection, it takes ten seconds and indicates with some sounds.

Broadband Connection

The term broadband commonly refers to high speed internet access that is always ON and faster than the traditional dial-up access. It uses a telephone line to connect to the internet. Broadband includes several high speed transmission technologies which are as follows:

1. Digital Subscriber Line (DSL)

DSL broadband connection provides internet access by transmitting digital data over wires of a local telephone networks. Some DSL form is :

High data rate Digital Subscriber Line (HDSL)

Very High data rate Digital Subscriber Line (VHDSL)

2. Cable Modem

Cable modem service enables cable operators to provide broadband using the same coaxial cable that deliver pictures and sound to your TV set.

3. Fiber Optic

Fiber optic technology converts electrical signals carrying data to light and sends the light through transparent glass fibers of about the diameter of human hair. It provide high data transmission speed than DSL and Cable Modem.

4. Broadband over Power Line (BPL)

BPL is the delivery of broadband over the existing low and medium voltage power distribution network. BPL speeds are comparable to DSL and cable modem speeds.

Wireless Connection

Wireless broadband can connect a home or business to the internet using a radio link between the customer's location and the service provider's facility. Wireless broadband can be mobile or fixed. Some way to connect the internet wirelessly are:

- **Wireless Fidelity (Wi-Fi)**

The wireless technology that utilizes radio frequencies to transfer data. Wi-Fi allows high speed internet connection without the use of wires.

- **Worldwide Interoperability for Microwave Access (WiMAX)**

WiMAX would operate similar to Wi-Fi but at higher speed, over greater distances and for a greater number of users

Integrated services digital network (ISDN)

It is a digital telephone service that can transmit voice, data and control information over an existing single telephone line. It was the first high speed alternative to analysis phone modems regularly. It is widely used for business purpose.

Web Browsing Software

A web browser or simply browser is a special software that enables the users to read/view web pages and jump from one web page to another. It is the software that is needed to find, retrieve, view and send information over the internet. Browsers are two types:

- **Text-Based Web Browsers** are the browser that support text only not graphics. E.g. Lynx
- **Graphical-Based Web Browsers** provide a GUI where the user can jump from one page to another by clicking on the hyperlink on a web page . E.g. Chrome, Mosaic, Internet Explorer etc.

The process of using browser to view information on the internet is known as Browsing or Surfing.

How is a server different from a workstation computer?

Server vs. Workstation.

A **server** is an application or device that performs service for connected clients as part of client **server** architecture. ...

A **workstation** is a personal **computer** that is used for high end applications such as graphic design, video editing, CAD, 3-D design, or other CPU and RAM intensive programs.

What is an example of a server?

There are many different types of **servers**, for **example**:

File server: a computer and storage device dedicated to storing files. ...

Print server: a computer that manages one or more printers,

network server: is a computer that manages network traffic.

Database server: a computer system that processes database queries.

What is the main purpose of a server?

- A server is a computer used in a network and which provides a service to a client. Servers usually have more processing **power**, memory and storage than client computers